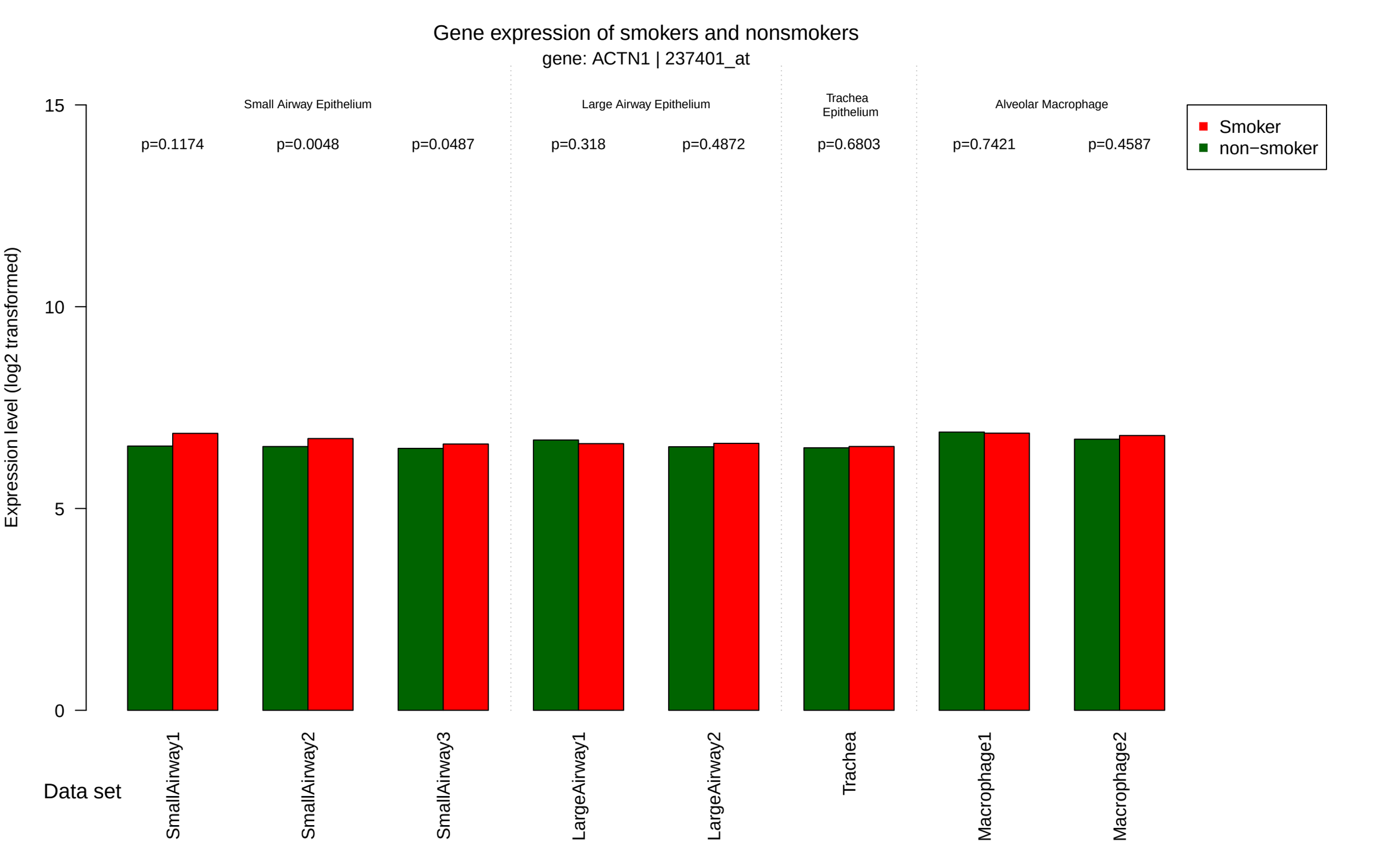
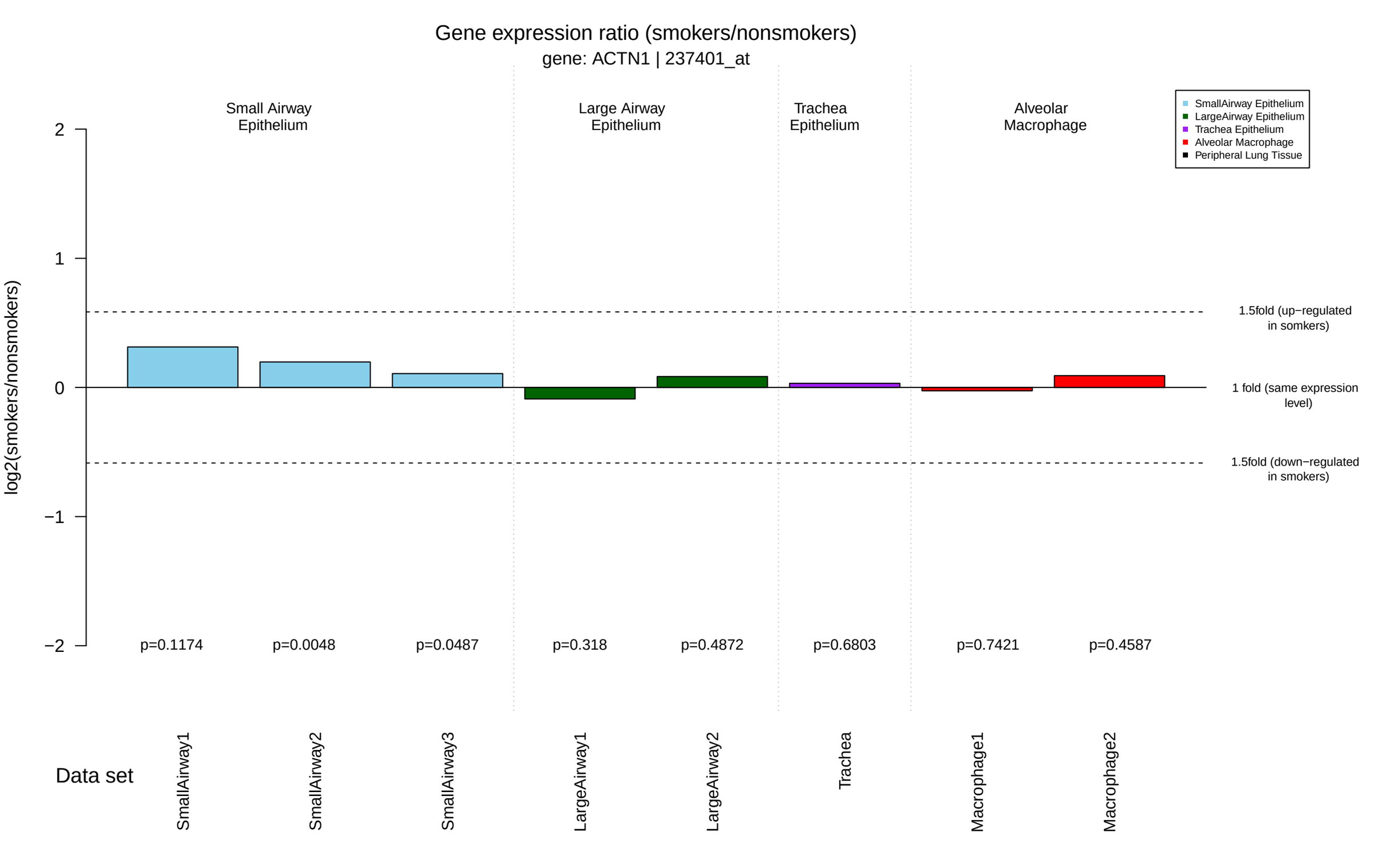
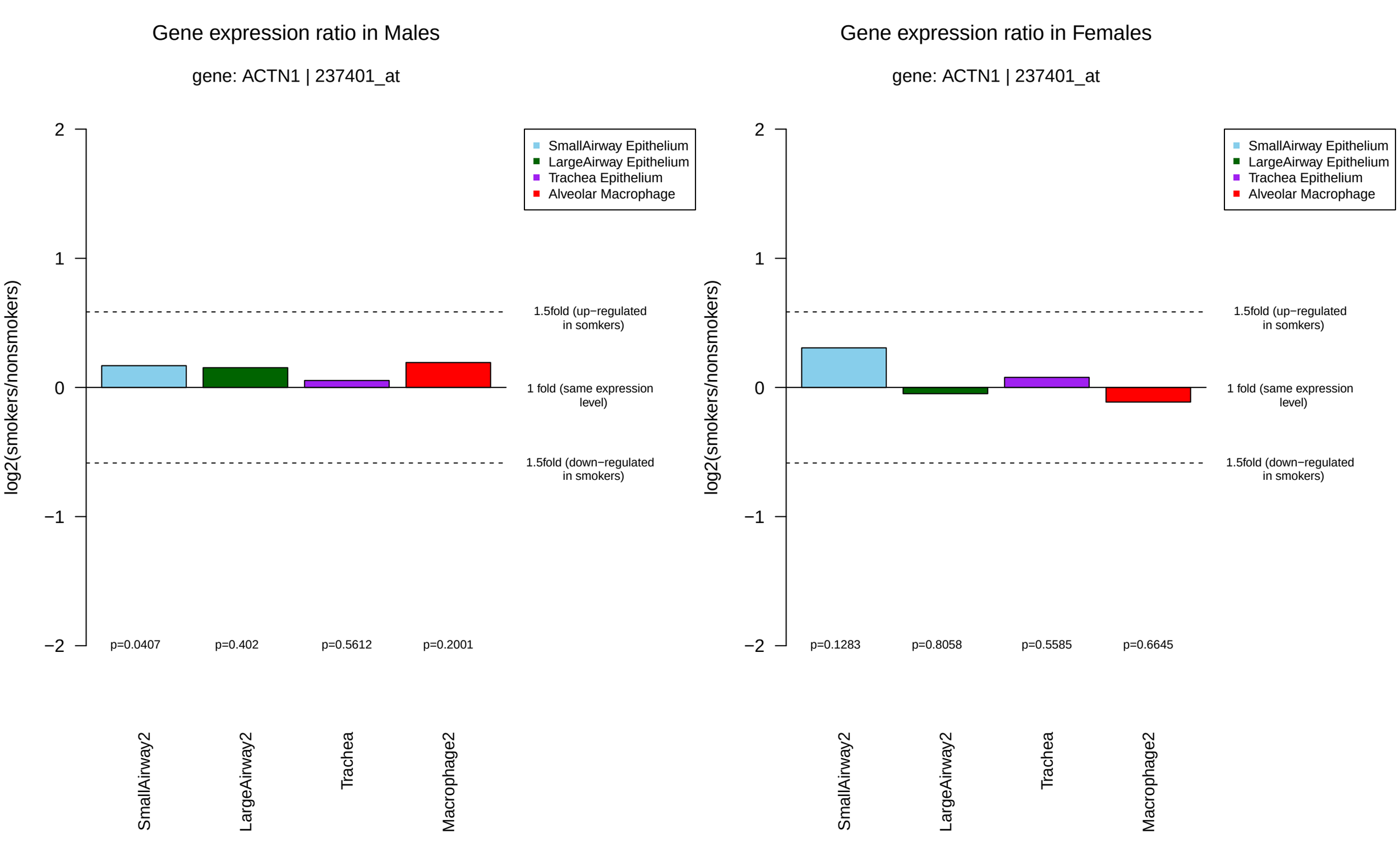
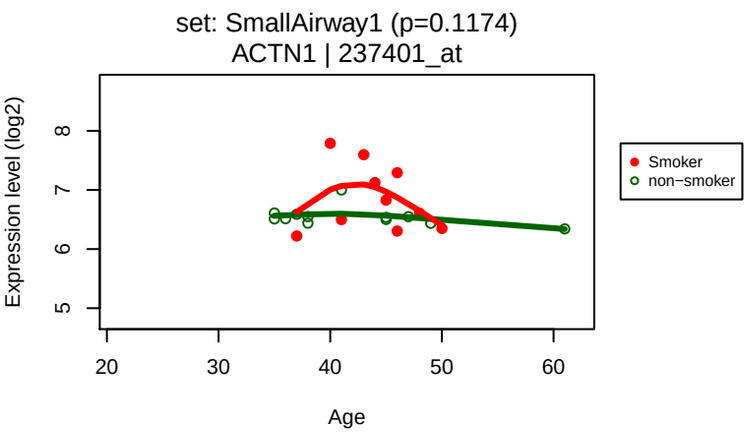
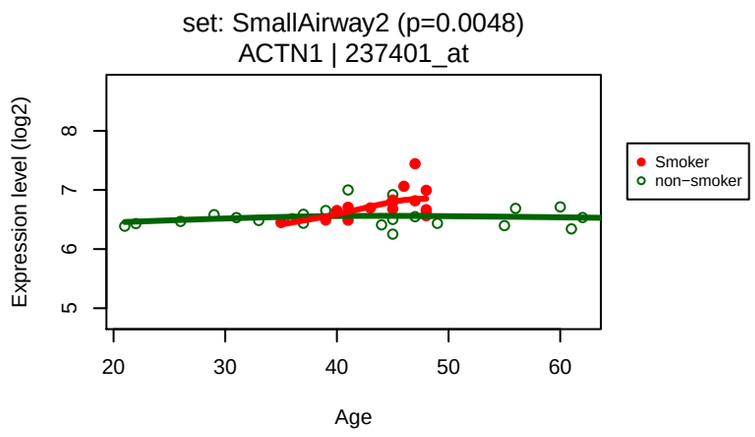
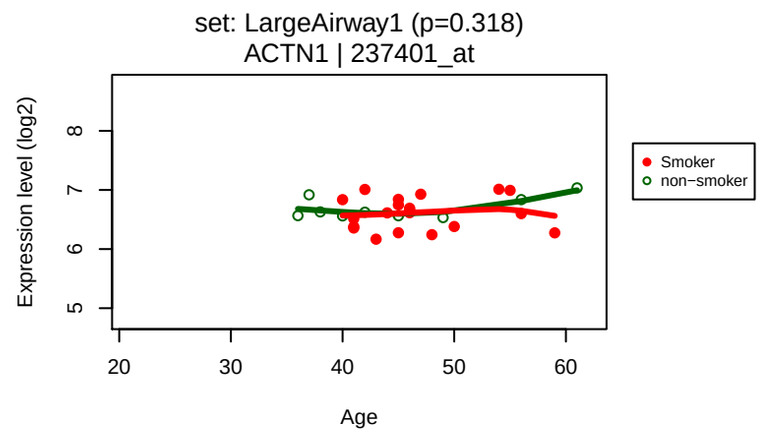

Gene expression of smokers and nonsmokers
gene: ACTN1 | 237401_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1174
p=0.0048
p=0.0487
p=0.318
p=0.4872
p=0.6803
p=0.7421
p=0.4587
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

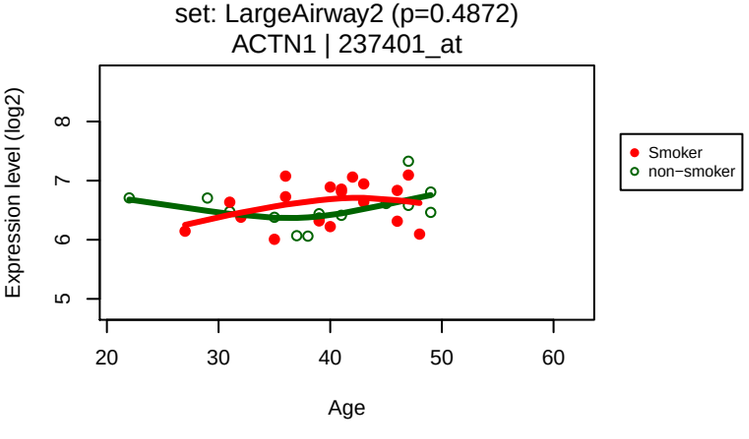

Gene expression ratio (smokers/nonsmokers)
gene: ACTN1 | 237401_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1174
p=0.0048
p=0.0487
p=0.318
p=0.4872
p=0.6803
p=0.7421
p=0.4587
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

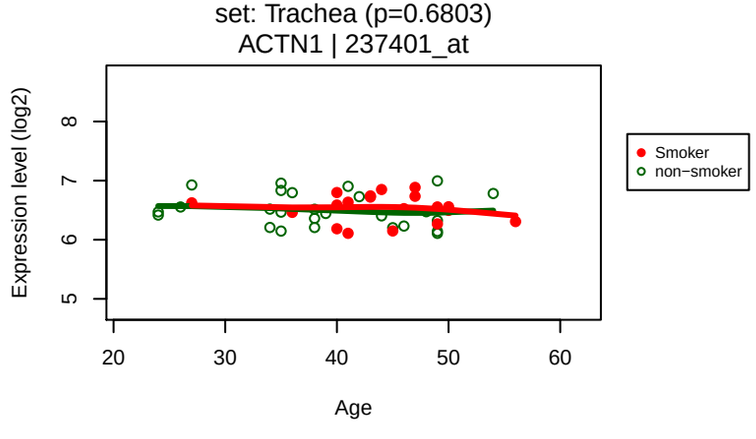

Gene expression ratio in Males
gene: ACTN1 | 237401_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0407
p=0.402
p=0.5612
p=0.2001
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ACTN1 | 237401_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1283
p=0.8058
p=0.5585
p=0.6645
SmallAirway2
LargeAirway2
Trachea
Macrophage2

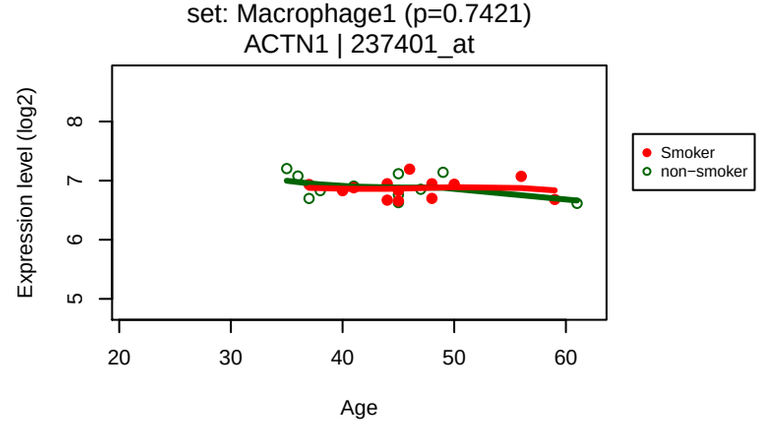

set: SmallAirway1 (p=0.1174)
ACTN1 | 237401_at
Expression level (log2)
Age
Smoker
non−smoker

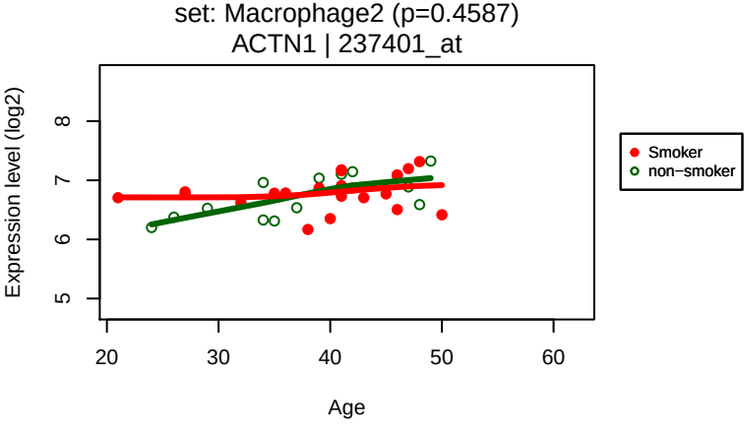

set: SmallAirway2 (p=0.0048)
ACTN1 | 237401_at
Expression level (log2)
Age
Smoker
non−smoker

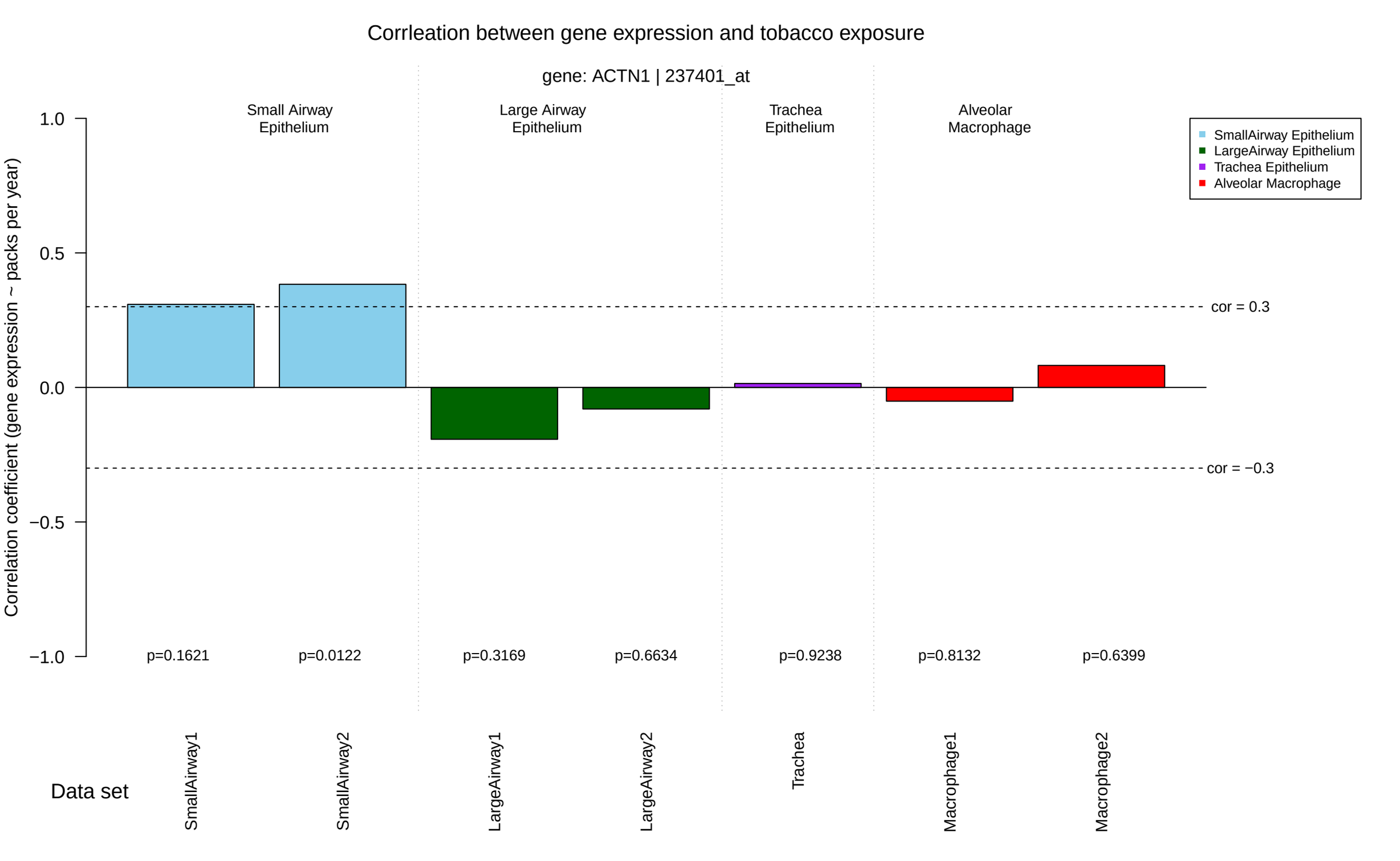

set: LargeAirway1 (p=0.318)
ACTN1 | 237401_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4872)
ACTN1 | 237401_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6803)
ACTN1 | 237401_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7421)
ACTN1 | 237401_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4587)
ACTN1 | 237401_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ACTN1 | 237401_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1621
p=0.0122
p=0.3169
p=0.6634
p=0.9238
p=0.8132
p=0.6399
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2